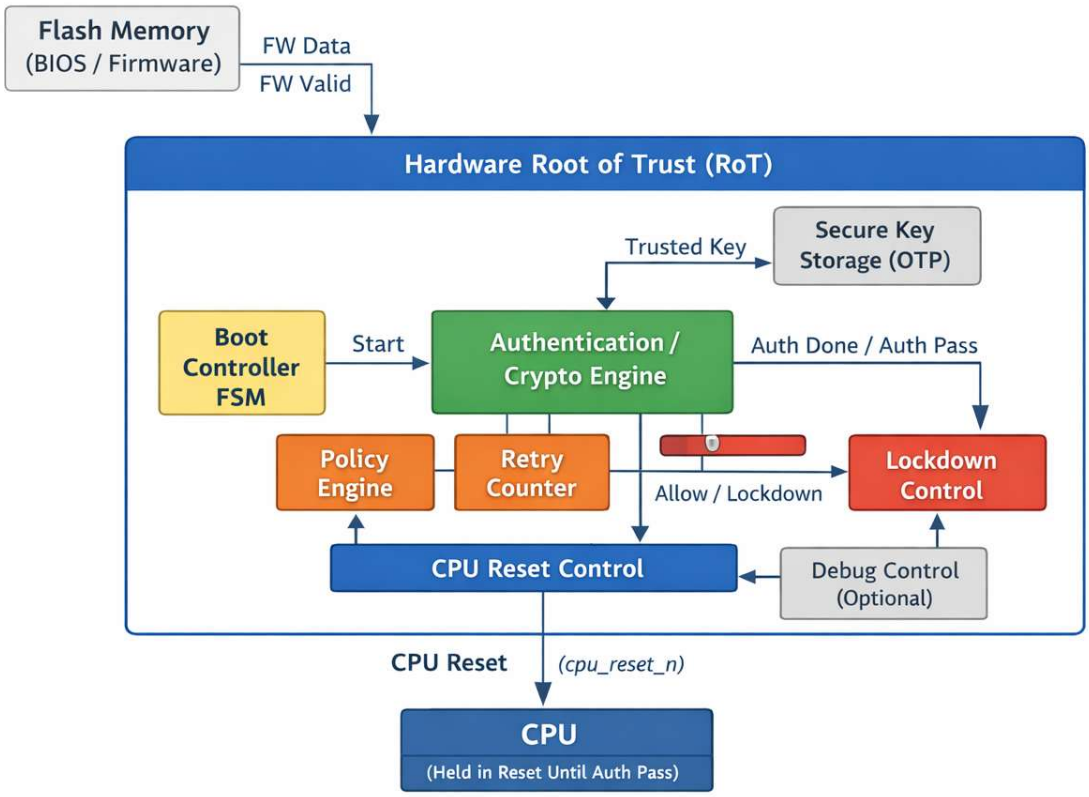
Hardware Root-of-Trust (RoT)

Block Diagram of a Hardware ROT :



Complete Module-Level and Pin-to-Pin Design Document

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1. Introduction

This document describes the complete **Hardware Root-of-Trust (RoT)** architecture implemented for **secure boot**, operating strictly **before CPU execution**. The RoT ensures that only authenticated firmware (BIOS/bootloader) stored in flash memory is allowed to execute by holding the CPU in reset until verification succeeds.

This design is:

- Hardware-enforced (no software dependency)
- Deterministic and fail-safe
- Suitable for SoC secure boot reference implementations

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2. Top-Level Module: rot_top

What it is

rot_top is the integration wrapper for the entire Root-of-Trust IP. It defines the **security boundary** of the RoT and exposes only minimal, safe control signals to the rest of the SoC.

Role

- Instantiates and connects all RoT submodules
- Routes firmware data and control signals
- Exposes CPU reset and boot status
- Prevents any direct external access to secure internals

Pin Description

Inputs

- clk : System clock used by RoT logic
- rst_n : Global reset (active low)
- boot_req : Indicates a boot request after power-on
- fw_data [N-1:0] : Firmware data stream from flash controller
- fw_valid : Indicates valid firmware data
- fw_last : Indicates last firmware word (optional)

Outputs

- cpu_reset_n : Controls CPU reset (active low)
- boot_done : Indicates boot decision completed
- boot_pass : Indicates firmware authentication success
- secure_mode : Indicates system is in secure boot mode
- debug_disable : Disables debug access during secure boot

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3. Boot Controller FSM: boot_ctrl_fsm

What it is

A finite state machine that controls the **entire secure boot flow**. It is the central decision coordinator but does **not** perform cryptography or access secret keys.

Role

- Enforces strict boot sequencing
- Triggers authentication
- Waits for authentication result
- Commands allow / deny / lockdown actions

FSM States

- FSM_RESET
- FSM_IDLE
- FSM_AUTH_START
- FSM_AUTH_WAIT
- FSM_CHECK_RESULT
- FSM_BOOT_ALLOW
- FSM_BOOT_DENY
- FSM_LOCKDOWN

Pin Description

Inputs

- clk
- rst_n

- boot_req
- auth_done
- auth_pass
- lockdown_req

Outputs

- start_auth : Triggers authentication engine
- boot_done
- boot_pass
- cpu_allow : Allows CPU reset release
- enter_lockdown

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4. Authentication Engine: auth_engine

What it is

A hardware authentication block that verifies firmware authenticity using a trusted reference from secure key storage.

Role

- Receives firmware data as a stream
- Computes hash or verifies signature (abstracted)
- Compares against trusted key/hash
- Reports pass/fail result

Pin Description

Inputs

- clk
- rst_n
- start_auth
- fw_data [N-1:0]
- fw_valid
- fw_last
- trusted_key [K-1:0]

Outputs

- auth_done : Authentication complete
- auth_pass : Authentication success

Notes

- Cryptographic strength is abstracted
- Focus is on functional correctness and control flow

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5. Secure Key Storage: secure_key_storage

What it is

A protected, immutable storage block holding **trusted root material** such as public keys or golden hashes.

Role

- Acts as trust anchor
- Prevents software or external access

- Supplies keys only to auth engine

Pin Description

Inputs

- clk
- rst_n

Outputs

- trusted_key [K-1:0]

Properties

- No write interface
- OTP / ROM modeled
- Only readable internally

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6. Firmware Data Interface: fw_data_if

What it is

Logical interface between external flash controller and RoT.

Role

- Supplies firmware data to authentication engine
- Decouples RoT from flash implementation

Signals

- fw_data
- fw_valid
- fw_last

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7. Policy Engine: policy_engine

What it is

A rule-evaluation block that decides final boot action based on authentication result and system policy.

Role

- Interprets auth results
- Applies retry limits
- Requests lockdown if needed

Pin Description

Inputs

- auth_done
- auth_pass
- retry_count

Outputs

- policy_allow
- policy_deny
- lockdown_req

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8. Retry Counter: retry_counter

What it is

A security counter that limits repeated authentication attempts.

Role

- Prevents brute-force attacks
- Triggers lockdown after threshold

Pin Description

Inputs

- clk
- rst_n
- auth_fail

Outputs

- retry_count
- retry_exceeded

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9. Lockdown Controller: lockdown_ctrl

What it is

A fail-safe security module that enforces permanent denial on critical failures.

Role

- Overrides all boot logic
- Forces CPU reset
- Disables debug permanently

Pin Description

Inputs

- lockdown_req

Outputs

- lockdown_active

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10. CPU Reset Control: cpu_reset_ctrl

What it is

Final enforcement point controlling CPU execution.

Role

- Holds CPU in reset until secure boot passes
- Reacts immediately to failures

Pin Description

Inputs

- cpu_allow
- lockdown_active

Outputs

- cpu_reset_n

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11. Debug Control (Optional): debug_ctrl

What it is

Security block that controls debug access during secure boot.

Role

- Prevents debug-based attacks
- Locks debug permanently on failure

Pin Description

Inputs

- secure_mode
- lockdown_active

Outputs

- debug_disable

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12. Summary

This RoT design:

- Operates strictly before CPU execution
- Enforces secure boot via hardware FSM
- Separates control, cryptography, and policy
- Is extensible and synthesizable

This document serves as a **complete design reference** for implementation, verification, and academic publication.